PHAZE

AI fitness web app with two phases: Bulk and Cut

Live: fitness-dual.vercel.app | Code: github.com/MananAli05/Fitness-Dual

1. Summary

PHAZE helps people build muscle (**Bulk**) or lose fat (**Cut**). After sign up, the app asks a few questions (gender, age, height, weight, activity, training level, goal) and calculates daily calories, protein, carbs, fat and water. It then gives a daily workout plan and a daily Pakistani meal plan, and tracks workouts, food, water and progress over time.

Area	Status
Screens from the scope (20)	19 built. The AI live camera workout is the one screen still pending.
Platform	Web app that works on desktop and mobile browsers (changed from React Native mobile app, as agreed).
Backend	Supabase (database, login, file storage, security). The Flask API from the scope was not needed.
AI service (FastAPI)	Not built yet. Planned next: pose detection, rep counting and form feedback.
Hosting	Frontend on Vercel. Database on Supabase.

2. Client requirements check

Compared against **Fitness_Dual_Project_Scope.pdf** (20 screens, Flask modules, FastAPI endpoints, database tables).

Screens

#	Scope screen	Status	Where in the app
1	Splash screen	Done	Animated BULK, CUT, EVOLVE intro, then login
2-3	Login and Sign up	Done	One page: existing email logs in, new email creates an account
4	Home dashboard	Done	Phase progress, today's workout, calories, protein, water, streak
5	Workout categories	Done	Workouts page with filters: Legs, Chest, Back, Shoulders, Arms, Core, Cardio
6	Exercise details	Done	Steps, muscles worked, common mistakes, sets and reps
7	Manual workout	Done	Live timer, log reps and kg per set, add or remove sets
8	AI workout (live camera)	Pending	Needs the AI service (MediaPipe pose detection)
9	Progress dashboard	Done	Charts for training minutes, volume, calories and water
10	Workout history	Done	List of past workouts
11	Nutrition dashboard	Done	Calories and macros eaten vs target, day picker
12	Meal plans	Done	7 day Pakistani meal plan, portions sized to the user's target
13	Water tracking	Done	Add or remove glasses, target from body weight
14-15	Profile and Edit profile	Done	One page: photo, body details, phase, live targets
16	Fitness goals	Done	Onboarding questions and phase selection

#	Scope screen	Status	Where in the app
17	Settings	Done	Notification switches, restart phase, change password
18	Workout summary	Done	Shown after finishing a workout
19	Notifications	Done	Meal, workout and water reminders, streak and phase updates
20	Help and support	Done	Questions and answers about the app

Backend, AI and database from the scope

Scope item	Status	What we did
Flask REST API (about 30 endpoints)	Changed	Supabase gives a secure API for every table. Rules in the database (Row Level Security) make sure each user only sees their own data, so a separate Flask server was not needed.
FastAPI AI service (9 endpoints)	Pending	Pose detection, rep counting, posture feedback. Planned for the next phase.
users, fitness_goals	Done	Stored in the **profiles** table (body details, goal, phase start date, photo).
workouts, workout_exercises	Done	**workouts** and **workout_sets** tables.
progress_records	Done	Calculated from saved workouts, food and water. No extra table needed.
meal_plans, food_logs	Done	**food_logs** table. Meal plans and the food list are kept in the app code.
water_logs, notifications, user_settings	Done	Tables with the same names.
exercises	Changed	17 exercises kept in the app code for now. Can move to a table later if an admin panel is needed.

From the project documentation (FR-01 to FR-06)

Requirement	Status	Note
FR-01 Account management	Done	Sign up, log in, log out, forgot password, email confirmation
FR-02 Profile and goal setting	Done	Age, height, weight, gender, level, goal, BMI
FR-03 Personalised workout plan	Done	Changes by weekday, goal and training level (rule based)
FR-04 Exercise tracking	Partly done	Manual sets, reps, timer and duration work. Automatic rep counting needs the AI part.
FR-05 Real time form feedback	Pending	Needs the AI camera part
FR-06 Progress tracking	Done	History, charts, streaks

3. Features and how they work

Feature	How it works
Sign up and login	User enters email and password. If the email is new, the app asks for a username (letters only) and a phase, then sends a branded confirmation email. A popup tells the user to check their inbox.
Onboarding	Seven short questions, one per screen. At the end the app shows the daily plan: calories, protein, carbs, fat, water and BMI.
Targets	Calories use the Mifflin St Jeor formula. Bulk adds 300 kcal, Cut removes 500 kcal. Protein is 1.8 g per kg on Bulk and 2.2 g per kg on Cut. Water is about 35 ml per kg.
Dashboard	Greeting, current phase and week, today's workout, and today's calories, protein, water and streak.
Workouts	A different plan each weekday (for example Push day, Leg day, Active recovery on Sunday). Beginners get easier exercises and fewer sets, advanced users get more sets. The user can also start an empty workout and add exercises.
Workout session	Timer, sets done and volume. The user types reps and kg and taps Log after each set. Finish saves the workout and opens a summary.
Progress and history	Charts for weekly training minutes, lifting volume, calories and water, plus the full workout history.
Nutrition	Search 49 common foods (mostly Pakistani, Roman Urdu also works, e.g. anda, chawal). Pick the amount and calories fill in automatically. Foods not in the list can be entered by hand. A daily meal plan with a Log button, water tracker, and a day picker to see earlier days.
Profile	Profile photo (JPG, PNG or WebP, max 2 MB), body details, phase, activity and level. Targets update live while editing.
Notifications and settings	Reminders for meals, workouts and water, plus phase and streak updates. Each type can be turned off. Users can restart their phase or change password.

4. Technology

Part	Technology	Why
Frontend	React 19 with Vite	Fast, widely used, easy for other developers to pick up
Styling	Tailwind CSS v4	One place for colours and spacing, consistent design
Animation	Framer Motion	Smooth page and splash animations
Navigation	React Router	Separate links for every page
Database and API	Supabase (PostgreSQL)	Real database with an automatic, secure API
Login	Supabase Auth	Email and password, confirmation and reset emails
File storage	Supabase Storage	Profile photos, 2 MB limit enforced on the server
Hosting	Vercel (frontend), Supabase (backend)	Automatic deploy on every push to GitHub
Planned AI	MediaPipe Pose in the browser	Pose detection on the device, no video sent to a server

5. Database

Seven tables plus one storage bucket. Every table has **Row Level Security**: a user can only read and change their own rows, even if someone tries to call the API directly.

Table	What it stores
profiles	One row per user: username, photo, goal, phase start date, gender, age, height, weight, activity, level, onboarding done
workouts	Each finished workout: title, goal, start and end time, duration
workout_sets	Each logged set: exercise, set number, reps, kg or seconds
food_logs	Each food eaten: date, meal, name, calories, protein, carbs, fat
water_logs	Glasses of water per user per day
user_settings	Which reminders are switched on
notifications	Reminders and updates, read or unread
avatars (storage)	Profile photos, one per user, JPG, PNG or WebP, max 2 MB

Database changes are kept as numbered SQL files in **supabase/migrations** (001 to 007). They are run in order in the Supabase SQL Editor. Branded email templates are in **supabase/templates**.

6. Code structure

All app code is in **frontend/src**. Each folder has one job, so it is easy to find things.

Folder	What is inside
pages/	One file per screen (Dashboard, Workouts, Nutrition, Profile...). Login and password reset are in pages/auth.
components/	Building blocks of the screens, grouped by feature (dashboard, workouts, nutrition...). Shared pieces like buttons and cards are in components/common.
context/	Data shared across screens: logged in user, profile, active workout, notifications
lib/	All talking to Supabase: login, saving workouts, food, water, settings, photos
utils/	Calculations: calorie targets, daily workout plan, meal plan portions, dates, validation
constants/	Fixed data: 17 exercises, 49 foods, weekly plans, menu items
routes/	List of pages and the rule that only logged in users can open the app
layouts/	The sidebar and mobile menu around app pages, and the split login layout

How data moves: a screen (pages) calls a function in lib, lib talks to Supabase, Supabase checks the security rules and saves or returns the data, and the screen updates.

7. Remaining work

Item	Notes
AI live camera workout	MediaPipe pose detection, rep counter, posture feedback (squat first)
Custom email sender	Use a custom domain with Resend or Brevo so emails come from PHAZE, not Gmail
Password change safety	Ask for the current password before changing it in Settings
Reminders when the app is closed	Today reminders appear inside the app. Email or push reminders need a scheduled job.
Exercise images or videos	Waiting for media from the client

8. How to present to the client

Suggested order (about 20 minutes)

- **Start with the goal** (1 min): one app for two phases, Bulk and Cut, personalised to the user.
- **Splash and sign up** (2 min): create a new account live, show the username step and the confirmation email.
- **Onboarding** (2 min): answer the 7 questions and show the calculated plan.
- **Dashboard** (2 min): phase card, today's workout, daily stats.
- **Workout** (4 min): open Workouts, show filters and an exercise detail, start today's plan, log two sets, finish, show the summary.
- **Nutrition** (3 min): search "roti", add 2 roti to lunch, log a meal from the plan, add water.
- **Progress** (2 min): charts and history update from what was just logged.
- **Profile and settings** (2 min): upload a photo, switch Bulk to Cut and watch targets change.
- **Mobile** (1 min): open the live link on a phone.
- **Requirements and next steps** (2 min): use section 2 of this document.

How to explain the code (if asked)

- Open GitHub and show the folder structure from section 6. Explain that each folder has one job.
- Open **utils/targets.js** to show the calorie formula, and **utils/workouts.js** for the daily plan.
- Open one migration file in **supabase/migrations** to show a table and its security rules.

Questions the client may ask

Question	Answer
Why web and not a mobile app?	Agreed change. It works on any phone browser, needs no app store approval, and one codebase covers desktop and mobile.
Where is the Flask backend?	Supabase replaces it: a secure database API with per user security rules. Less code to maintain and nothing extra to host.
When will the AI camera work?	It is the next phase: pose detection with MediaPipe, rep counting and form feedback for 5 to 7 webcam friendly exercises.
Is user data safe?	Yes. Every table has row level security, passwords are handled by Supabase Auth, and photos are limited to 2 MB image files.
Can we add more foods or exercises?	Yes. They are in constants/foods.js and constants/exercises.js. An admin panel can be added later.